Prediction of Post-Fire Tree Mortality Using Normalized Burn Ratio (NBR) and Palmer Drought Severity Index (PDSI) Integration: A Machine Learning Approach

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Abstract

The devastating consequences of wildfires extend beyond immediate tree loss, impacting forest ecosystems for years. This study delves into using remote sensing and machine learning to forecast post-fire tree mortality by examining the relation between burn severity, water stress, and delayed mortality.

The investigation analyzed Normalized Burn Ratio (NBR) and Palmer Drought Severity Index (PDSI) in relation to tree mortality data. While no strong correlations were found, subtle connections hint at intricate environmental factor relationships.

Machine learning models, trained on combined NBR and PDSI data, accurately predicted tree mortality. The Random Forest model notably achieved 94.96% accuracy across tree categorizations, showcasing its effectiveness.

Feature selection, utilizing long-term differenced NBR, Relative differenced NBR measures, and multi-year PDSI data, enabled the model to precisely predict delayed mortality, emphasizing its importance.

These findings advocate for using remote sensing and machine learning in post-fire forest management. Identifying areas susceptible to delayed mortality aids resource allocation, mitigating ecological and economic losses.

Further validation on independent datasets and diverse data sources is required, along with advanced models, to refine predictions and enhance comprehension of post-fire recovery dynamics.

This study foresees a future in forest management by actively predicting delayed tree mortality, aiming for resilient ecosystems amidst frequent and severe wildfires.

Introduction

Post-fire tree mortality is a pressing ecological concern, representing a substantial proportion of tree loss in severely burned landscapes. Studies estimate this mortality to account for 40 to 60 percent of overall tree loss (Cansler & McKenzie, 2012), significantly disrupting ecosystem functioning. This loss has been associated with a potential reduction of up to 30% in carbon sequestration capacity within affected regions (Mo et al., 2023).

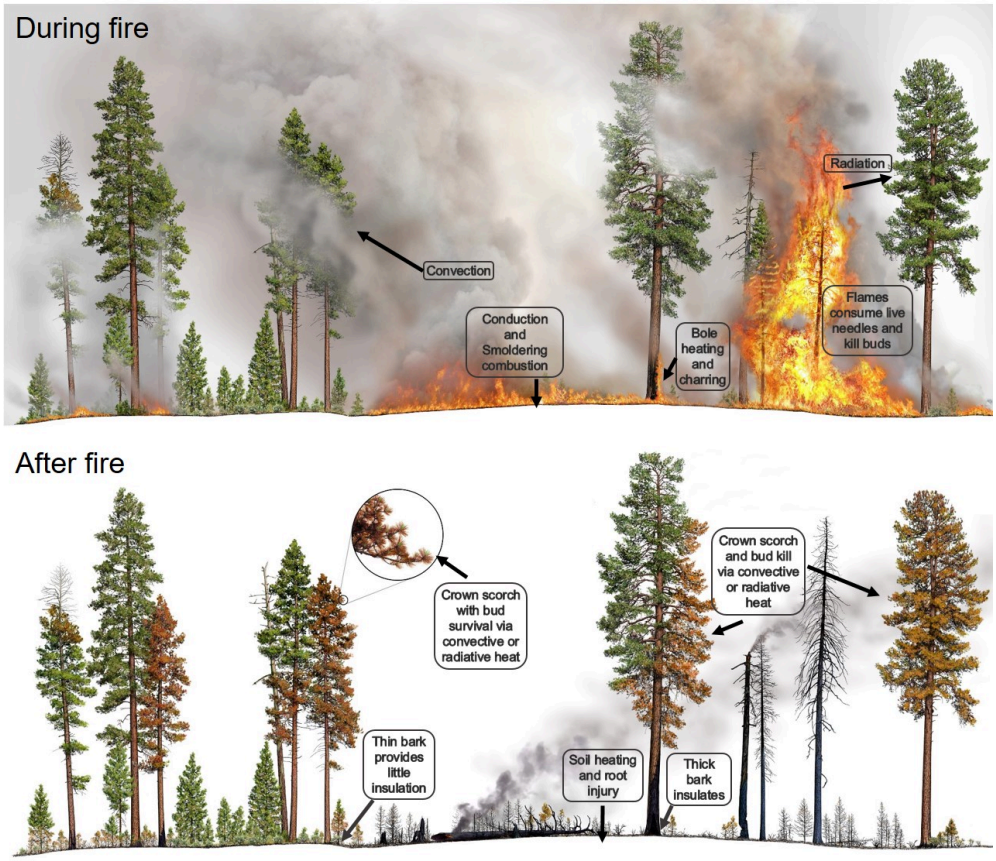


Figure 1: Impacts of wildfires on trees (Credit: Hood et al. 2018)

Preserving mature trees is crucial for optimal carbon absorption. Research conducted by the Urban Forestry Network indicates that while young trees absorb 13 pounds of CO_2 annually, mature trees capture an impressive 48 pounds per year by the age of 10 (Urban Forestry Network, n.d.). Protecting and nurturing mature trees alongside tree planting initiatives is essential to maximize CO_2 uptake, contributing significantly to mitigating the effects of climate change.

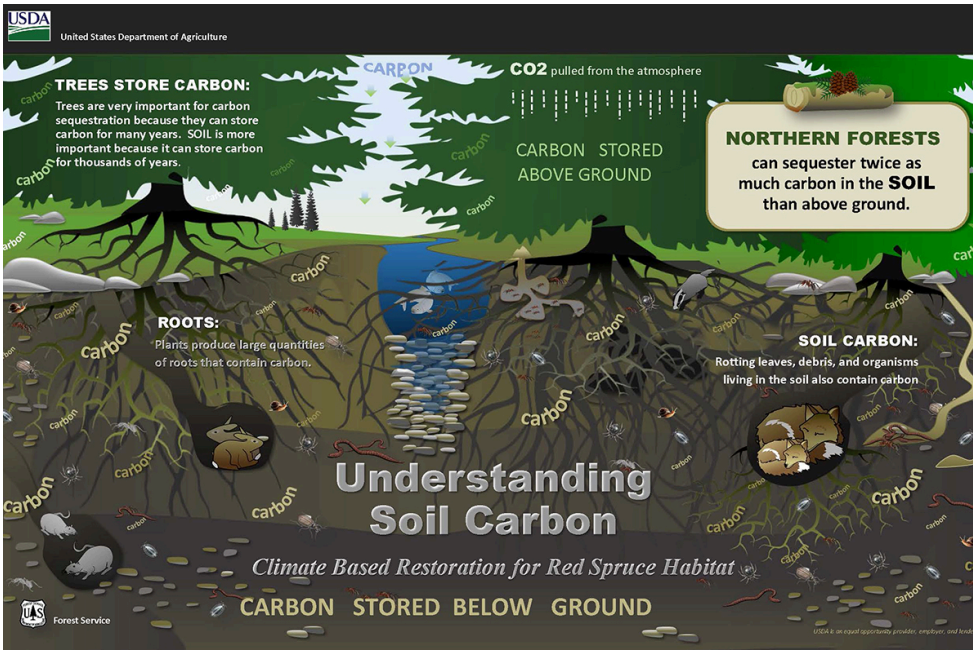


Figure 2: Mature forests, containing 70% of soil carbon, risk a positive feedback loop: carbon release accelerates climate change, reducing forest cover and exacerbating carbon loss. (Credit: USDA and USFS)

Mature forests play a vital role in ecosystem stability and carbon storage, containing a significant portion of soil carbon essential for climate regulation. They represent an integral aspect of Western landscapes, accumulating generations of biomass crucial for ecosystem resilience. These forests store up to 70% of the carbon in forest ecosystems, including deep soil carbon sequestered over centuries (Brack, 2019). However, wildfires pose a considerable threat by potentially releasing substantial amounts of this deep soil carbon, accelerating climate change, and disrupting soil ecology (Santín et al., 2015). The loss of mature trees in wildfires can lead to the release of stored carbon, exacerbating these environmental impacts.

The economic implications of post-fire tree mortality are substantial, with landscape restoration costs in the United States exceeding millions of dollars. These expenses strain forestry management budgets, particularly as wildfires' frequency and severity continue to rise (Thomas et al., 2017).

Beyond direct financial burdens, post-fire tree mortality significantly affects wildlife habitat quality (Rockweit et al., 2017) and alters hydrological processes, resulting in reduced water yield. Research indicates a 20 percent or more decline in annual water yield in affected landscapes (Lane et al., 2010), exacerbating challenges related to water availability and quality.

Current approaches often focus solely on assessing fire severity or drought conditions, neglecting the compounded impact of their interaction on tree survival post-wildfires (Sieg et al., 2006). Additionally, contemporary research predominantly aims at predicting wildfire occurrences without fully addressing the complexities of post-fire forest recovery dynamics, signaling a necessary shift towards a more comprehensive and holistic approach (Liu et al., 2020).

I address the connection between fire and its enduring impact on trees through the analysis of remote sensing data. This approach enables the mapping of burn severity across expansive areas with unparalleled accuracy, surpassing the constraints of ground-based observations. Key to my methodology are the utilization of the Normalized Burn Ratio (NBR) (Key & Benson, 2006) and the Palmer Drought Severity Index (PDSI) (Palmer, 1965).

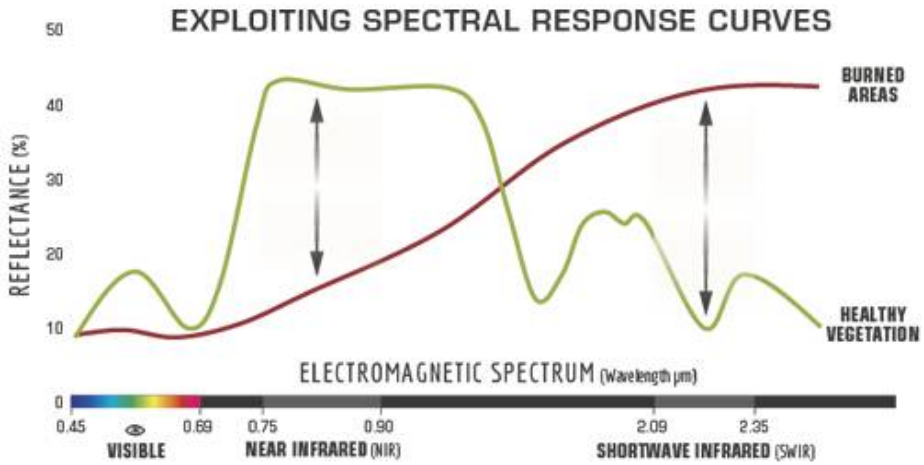


Figure 3: Comparison of the spectral response of healthy vegetation and burned areas (Credit: USDA Forest Service)

The NBR functions as a gauge of fire severity, contrasting the reflection of near-infrared light (NIR) by healthy vegetation with the diminished NIR reflectance in burned areas (Key & Benson, 2006). This comparison allows for precise identification of fire impact.

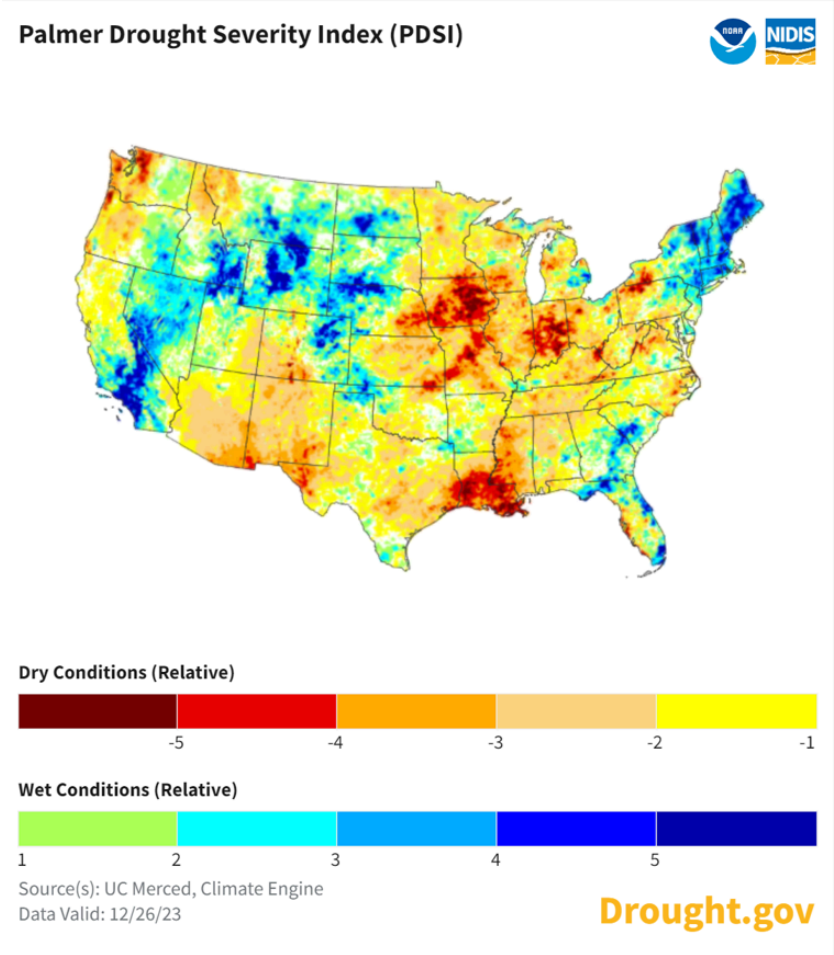


Figure 4: PDSI data for Contiguous U.S. as of 12/21/2023 (Credit: drought.gov)

Meanwhile, the PDSI offers a broader perspective by uncovering the prolonged influence of drought on tree survival. By integrating factors such as precipitation, temperature, and soil moisture, the PDSI unveils whether dry conditions exacerbate the effects of fire.

This project aimed to address the following questions:
How can we accurately predict and mitigate the significant (40-60%) post-fire tree mortality, which disrupts ecosystem functioning and reduces carbon sequestration capacity by up to 30%?
Can remote sensing data predict delayed mortality following wildfires?
Is there a correlation between post-fire tree mortality, burn severity assessed via Normalized Burn Ratio (NBR), and water stress indirectly measured using the Palmer Drought Severity Index (PDSI)?

There were two primary objectives:

- Investigate the relationship between post-fire tree mortality, burn severity (NBR), and water stress (PDSI) to ascertain potential correlations.
- Utilize remote sensing data incorporating NBR and PDSI to construct a predictive model for anticipating delayed post-fire tree mortality and promptly identify priority areas for comprehensive field studies, aiding in targeted post-fire management strategies.

Methodology

This research employed a multi-faceted approach, integrating diverse datasets and leveraging remote sensing techniques to explain the intricate relationships between drought, wildfire severity, and vegetation dynamics across a range of spatial and temporal scales. Key steps included data acquisition from various sources, rigorous spatial and temporal alignment, derivation of pertinent indices, and strategic data merging to construct a comprehensive framework for subsequent analyses.

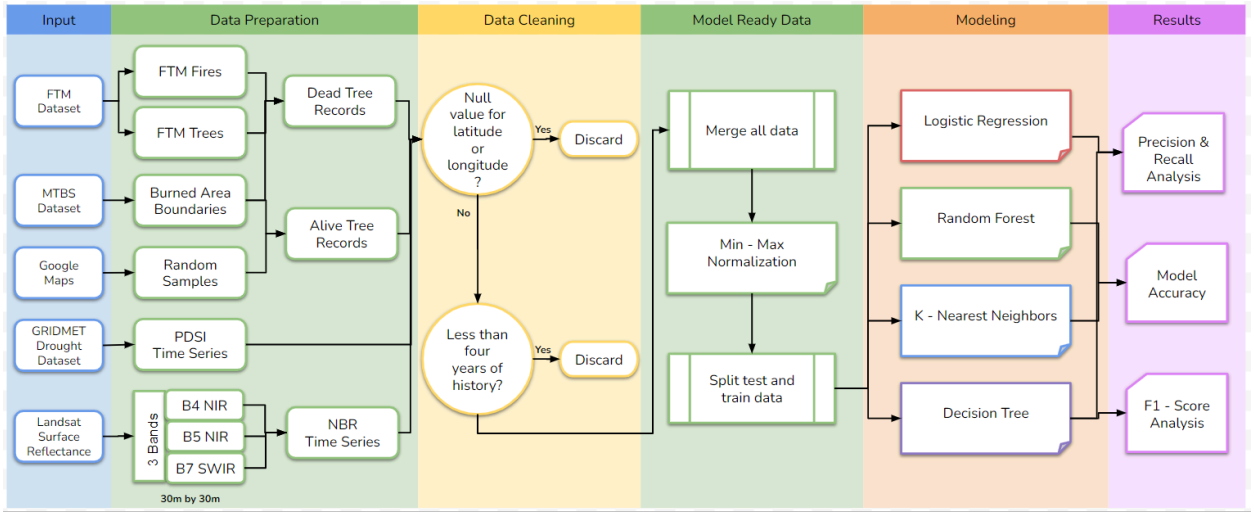


Figure 5: An overview of the methodology used in this project is depicted in the diagram above. (Credit: Reva Khaire)

Data Processing

1. Accessing Fire and Tree Mortality (FTM) Database:
    - Downloaded FTM data from the USDA website to ensure access to the most current and reliable information.
    - Consulted accompanying metadata to gain a thorough understanding of the data structure, including the layout of the "Fires" and "Trees" datasets and the meaning of their respective attributes.
    - Extracted comprehensive historical wildfire records and tree mortality data, focusing on:
        - Fire Year
        - Geographical coordinates (latitude and longitude)
        - Associated mortality rates
    - Performed a data cleaning step to remove any fire records with missing coordinates, ensuring the integrity and accuracy of spatial analyses.

2. Retrieving Burned Area Boundaries from Monitoring Trends in Burn Severity (MTBS) Dataset:
    - Accessed the MTBS dataset through Google Earth Engine (GEE) for its robust spatial capabilities.

- Utilized MTBS to acquire precise burned area boundaries corresponding to the fire incidents documented within the FTM database.
- This eliminated the need for manual delineation, ensuring consistency and efficiency in data retrieval.

3. Selection of Random Points within Burned Areas:
    - Employed a randomized selection process to designate a representative sample of 10 points within each identified burned area, as outlined by the MTBS dataset.
    - Collected the latitude and longitude coordinates of these 10 sample points, serving as essential spatial references for subsequent data retrieval.
    - These meticulously selected points, with their precise coordinates, formed foundational anchors for the subsequent collection of PDSI and surface reflectance data.

4. PDSI Data Collection:
    - Import Necessary Libraries: This step involves loading the Earth Engine Python API to interact with Earth Engine datasets.
    - Access PDSI Data: This step fetches the "GRIDMET DROUGHT: CONUS Drought Indices" dataset, specifically accessing the band containing PDSI values.
    - Filter for 4-Year Time Window:
        - It retrieves the "year of fire" information for each sample point from your data.
        - It creates a 4-year time window, starting from the year before the fire and extending to three years after the fire.
        - It filters the PDSI data to include only this specific time period for each sample point.
    - Extract Time Series for Each Sample Point:
        - It loops through each sample point's coordinates (longitude and latitude).
        - It extracts the PDSI time series data for that specific point within the defined 4-year window, using the coordinates as a spatial filter.

5. NBR Data Collection:
    - Import Landsat Surface Reflectance: This step imports the Landsat surface reflectance dataset:
        - TM (Thematic Mapper): 1982-2012
        - ETM+ (Enhanced Thematic Mapper Plus): 1999-present (with partial failure in 2003)
        - OLI (Operational Land Imager): 2013-present
    - By incorporating data from all three sensors, we can create an NBR time series that extends back to the early 1980s, providing a comprehensive historical record of burn severity.
    - Filter for 4-Year Time Window and Select Bands:
        - It applies the same 4-year time window filter used for PDSI data.
        - It selects three specific bands:
            - B4: Red, sensitive to chlorophyll absorption (NIR)
            - B5: Near-Infrared (NIR), sensitive to vegetation health

        - B7: Short-Wave Infrared (SWIR), sensitive to moisture content
    - Calculate NBR:
        - It calculates the Normalized Burn Ratio (NBR) using the formula: (NIR - SWIR2) / (NIR + SWIR2). This index highlights burned areas due to their distinct spectral characteristics.
    - Mask Clouds and Cloud Shadows:
        - It assumes cloud and cloud shadow masks are available within the dataset.
        - It applies these masks to exclude pixels affected by clouds or cloud shadows, ensuring accurate NBR values.
    - Apply Scaling Factors:
        - It applies appropriate scaling factors to ensure consistency with standard NBR values.
    - Extract Time Series for Each Sample Point:
        - It extracts the NBR time series for each sample point using their coordinates, similar to the PDSI extraction process.
    - Compute Yearly and Monthly Means:
        - It calculates the average NBR values for each year and month within the time series, providing aggregated insights into NBR trends over time.
    - Data Exclusion:
        - It excludes any records where the complete 4-year time series of PDSI or NBR data is not available, ensuring data quality and consistency for subsequent analyses.

6. Data Collection at Unburned (Living) Areas:
    - Selection of Random Points within Unburned Areas:
        - Employed a randomized selection process to designate a representative sample of points within unburned areas.
        - Critically, to maintain data consistency, these unburned areas were meticulously selected from the same states where fires were listed in the FTM database.
        - This ensured a comparable spatial context for subsequent analyses.Collected the latitude and longitude coordinates of these sample points for PDSI and NBR data collection.
    - PDSI Data Collection:
        - Filter for 4-Year Time Window:
            - Utilized the "year of fire" information from the FTM database to create a matching 4-year time window for each unburned point, ensuring temporal consistency with burned area data.
        - Extract Time Series for Each Sample Point:
            - Extracted the PDSI time series data for each unburned point within the defined 4-year window, using their coordinates as spatial filters.
    - NBR Data Collection:
        - Filter for 4-Year Time Window and Select Bands:
            - Applied the same 4-year time window filter used for burned area NBR data, maintaining temporal alignment.

        - Selected the three relevant bands (B5, B4, B7) for NBR calculation.
    - Calculate NBR:
        - Calculated NBR values for each image in the time series using the established formula.
    - Mask Clouds and Cloud Shadows:
        - Applied available cloud and cloud shadow masks to exclude affected pixels and ensure accurate NBR values.
    - Apply Scaling Factors:
        - Applied appropriate scaling factors (0.0001) to align with standard NBR values.
    - Extract Time Series for Each Sample Point:
        - Extracted the NBR time series for each unburned point using their coordinates.
    - Compute Yearly and Monthly Means:
        - Calculated average NBR values for each year and month, providing aggregated insights into trends over time.
- Data Exclusion:
    - Excluded records with incomplete 4-year time series for PDSI or NBR data, maintaining data quality and consistency.

7. 180-Day Rolling Mean for Medium-Term Seasonality:
    - Calculate 180-Day Rolling Mean for Each Row:
        - For each row in the dataset (representing a specific date), compute the 180-day rolling mean for both PDSI and NBR values.
        - This smooths out short-term fluctuations and highlights medium-term trends within the 180-day window.
    - Average Rolling Means for 6-Month Periods:
        - Divide the 180-day rolling mean values into 4-year * 2 periods (8 periods in total), each representing 6 months.
        - Calculate the average rolling mean value for each 6-month period.
    - This creates 8 new attributes for PDSI, 2 for each year thus 8 attributes for 4 years. Similarly it creates 8 new attributes for NBR.

8. 2-Year Rolling Mean for Long-Term Patterns:
    - Calculate 2-Year Rolling Mean for Each Row:
        - For each row, compute the 2-year rolling mean for both PDSI and NBR, aggregating values across consecutive years.
    - Average Rolling Means for Each Year:
        - Calculate the average 2-year rolling mean value for each year within the 4-year period.
    - This creates 4 new attributes for PDSI, 1 for each year thus 4 attributes for 4 years. Similarly it creates 4 new attributes for NBR.

Benefits of Additional Attributes for Capturing Seasonality in Predictive Models:

- Highlight Medium-Term Trends: The 180-day rolling mean attributes effectively capture medium-term seasonal patterns within each year, which can play a crucial role in wildfire behavior and outcomes.
- Reveal Long-Term Weather Patterns: The 2-year rolling mean attributes capture overall average conditions and long-term weather patterns, potentially influencing fire severity and post-fire recovery trajectories.
- Improve Model Performance: By incorporating these attributes, predictive models can better account for the complex relationships between drought, vegetation conditions, seasonal patterns, and wildfire dynamics, potentially leading to more accurate predictions and insights.

9. Quantifying Burn Severity: dNBR and RdNBR Calculation
    - To assess the magnitude and severity of burn effects within the affected areas, two metrics were derived: differenced Normalized Burn Ratio (dNBR) and Relative differenced Normalized Burn Ratio (RdNBR).
    - dNBR: Calculated as the post-fire NBR minus the pre-fire NBR for each individual image date within the time series, providing a direct measure of change in spectral response due to burning.
    - RdNBR: Normalized dNBR by dividing it by the pre-fire NBR, accounting for inherent variability in vegetation conditions across different locations. This offers a more standardized measure of burn severity relative to baseline reflectance characteristics.
    - These metrics allow for quantitative evaluation of fire impacts on surface reflectance and, consequently, vegetation health and burn severity across the landscape.
    - dNBR and RdNBR were calculated separately for medium and long term NBR values.

10. Data Integration: Constituting a Cohesive Dataset
    - PDSI and NBR data for both burned and unburned areas were consolidated into a comprehensive dataset, facilitating subsequent analyses and comparisons.
    - This involved spatial and temporal alignment of data from various sources, ensuring consistency and compatibility for integrated analyses.
    - The merged dataset encompasses a unified representation of drought conditions (PDSI) and burn severity (NBR) patterns across both affected and unaffected areas, enabling comprehensive investigations into environmental drivers and consequences of wildfire events.
    - This combined data framework fosters robust statistical analyses, model development, and exploration of interrelationships between drought, fire, and vegetation dynamics.

Model Development

- Feature Selection:
  An iterative approach employing multiple trial runs with various feature combinations was used to identify the most influential features for tree mortality. This method facilitated the exploration of diverse temporal scales and feature interactions. The final feature set was chosen to capture both multi-temporal dynamics of water stress (PDSI annualized mean values for 4 years starting

with the year before the year of fire) and long-term variations in burn severity (long-term dNBR and long-term RdNBR).

- Model Training and Evaluation:
  Four machine learning algorithms were utilized for prediction: SVC, Random Forest, Decision Tree, and K-Nearest Neighbors. Each model offers varying strengths in handling complex relationships and non-linear patterns within data. A single train-test split methodology was adopted for model development, ensuring efficient data utilization and robust performance assessment.

- Performance Metrics:
  Accuracy, precision, recall, and F1-score were employed to evaluate model performance. These metrics provide a comprehensive assessment of the ability to correctly identify different classes of tree mortality (alive, immediate, and delayed), ensuring reliable model selection for further analysis.

Results

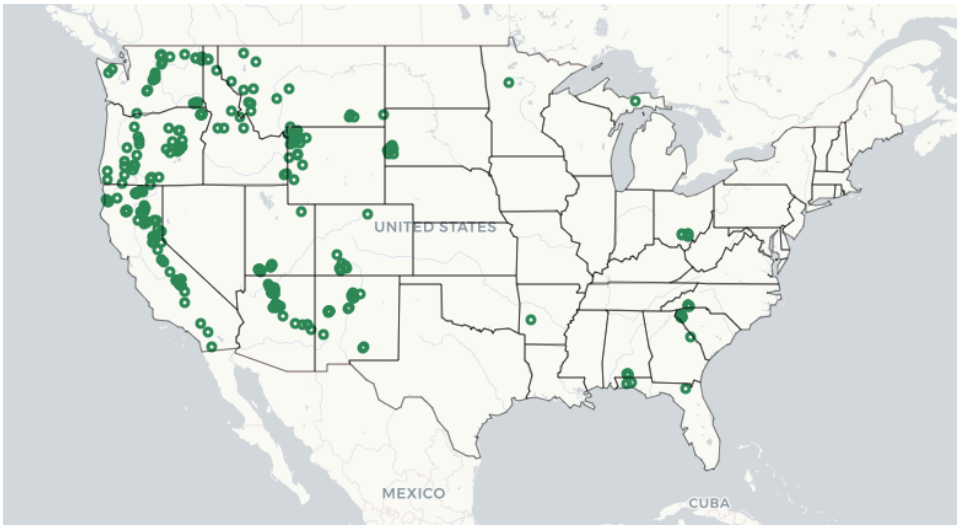


Figure 6: Wildfire Locations in the FTM database.(Credit: Reva Khaire)

Fire records within the Fire and Tree Mortality (FTM) database pertaining to the following contiguous and non-contiguous U.S. states were utilized for this research: Washington, Oregon, Colorado, Wyoming, Ohio, Alabama, South Carolina, Arizona, California, New Mexico, Utah, Idaho, Montana, Arkansas, North Carolina, Michigan, Florida, Alaska, Georgia, South Dakota, and Minnesota.

Long-Term and Short-Term Burn Severity Analysis in Fire-impacted and Living Trees

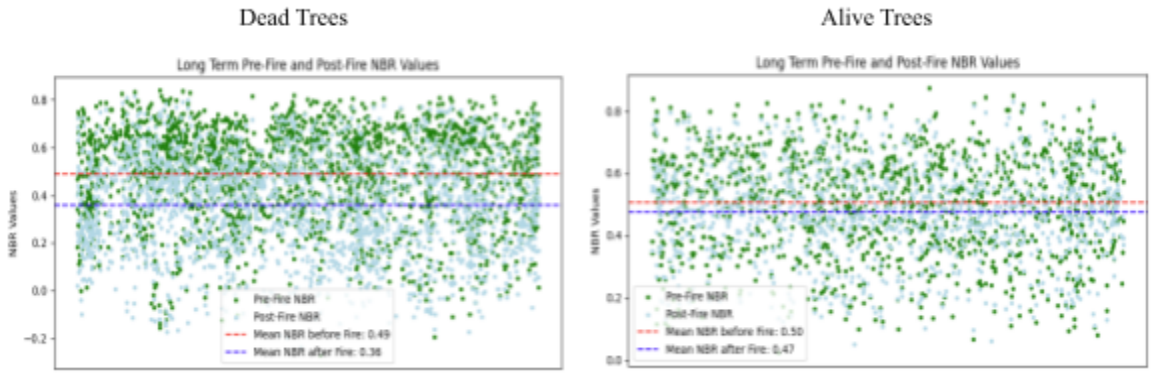


Figure 7: Comparison of NBR Long-term mean values for fire-impacted trees (left) and living trees (right) with notable drop in NBR values post fire for fire-impacted trees on left. (Credit: Reva Khaire)

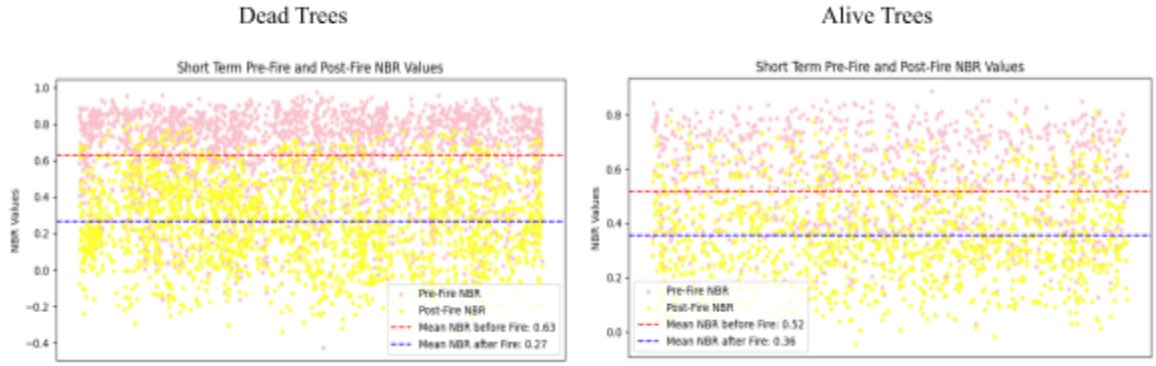


Figure 8: Comparison of NBR Short-term mean values for fire-impacted trees (left) and living trees (right) with notable drop in NBR values post fire for fire-impacted trees on left. (Credit: Reva Khaire)

	Fire-impacted Trees			Living Trees		
NBR Trend	Pre Fire	Post Fire	Difference	*Pre Fire	*Post Fire	Difference
Long Term	0.49	0.36	0.13	0.50	0.47	0.03
Short Term	0.63	0.27	0.36	0.52	0.36	0.16

Table 1: Comparison of long term and short term NBR values between fire-impacted and living trees (Credit: Reva Khaire)

Fire-impacted trees displayed a significantly larger and persistent NBR decline compared to live trees, suggestive of greater burn severity and extensive fire-induced thermal alterations. While both exhibited NBR drops, the effect was more pronounced in fire-impacted trees, likely due to enhanced fuel dryness or altered canopy structure. Live tree NBR changes, however, may primarily reflect natural seasonal variations, as their "Pre/Post Fire" designation serves for comparison with deceased NBR values.

Four-Year NBR and PDSI Trends in Fire-impacted Forests Beginning a Year Before Fire:

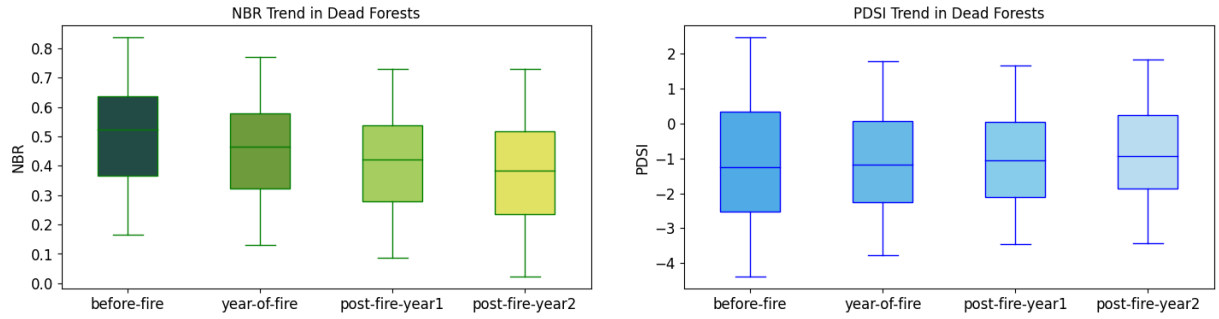


Figure 9: Four-year NBR and PDSI Trends in fire-impacted forests beginning a year before fire. (Credit: Reva Khaire)

A consistent decline in Normalized Burn Ratio (NBR) coincided with a progressive rise in Palmer Drought Severity Index (PDSI) across a four-year period from year before the wildfire detection in the affected forests.

Analysis of Post-Fire Water Stress:

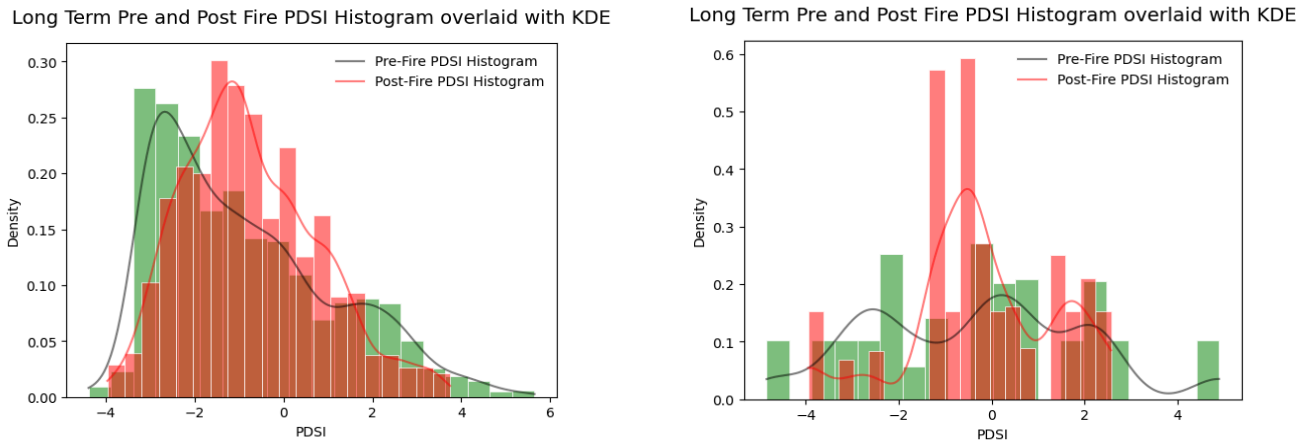


Figure 10: Comparison of short term PDSI values between fire-impacted and living trees (Credit: Reva Khaire)

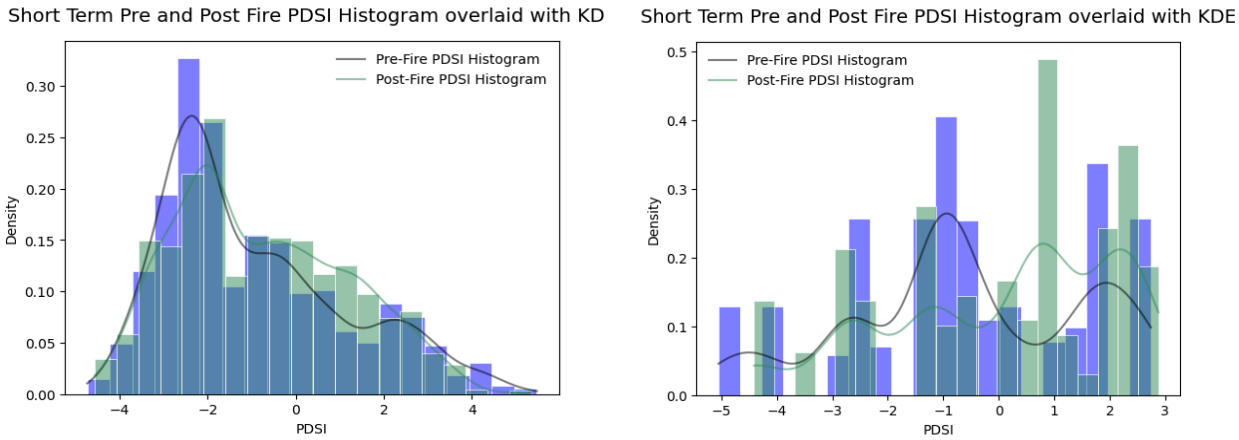


Figure 11: Comparison of short term PDSI values between fire-impacted and living trees (Credit: Reva Khaire)

Examination of the long-term and short-term Palmer Drought Severity Index (PDSI) histograms revealed a contrasting pattern between trees impacted by wildfire and those not impacted. For fire-affected trees, a noticeable increase in PDSI values was observed across both temporal scales. This suggests that post-fire conditions are characterized by heightened water stress compared to the pre-fire period. This is likely due to a combination of factors, including disrupted hydrological cycles, reduced canopy cover leading to increased evapotranspiration, and potential damage to root systems affecting water uptake. Conversely, the PDSI histograms for trees not impacted by wildfires did not exhibit a significant shift, indicating they remained within their pre-fire water stress regimes.

Correlation Chart

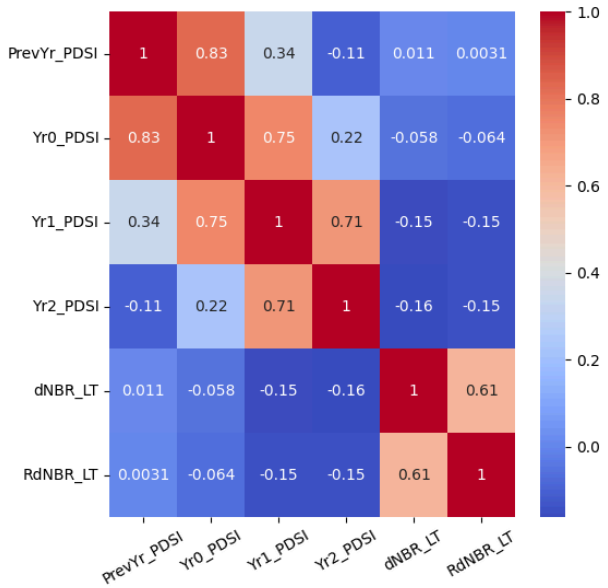


Figure 12: **Correlation Matrix for the attributes used in models** ***(Credit: Reva Khaire)***

Drought conditions showed strong persistence, with correlations between consecutive years reaching as high as 0.83. Moderate influence of prior year's drought on subsequent years was observed (correlations around 0.34 and 0.22). Direct links between pre-fire drought and burn severity were weak, with correlations ranging from -0.15 to -0.05. Potential delayed effects of drought on vegetation recovery and burn severity emerged, with correlations between post-fire drought and burn severity indices reaching -0.16. The strong correlation between the 4 years of PDSI data underscores the persistence of drought's impacts on the landscape. This reinforces the importance of considering drought's long-term effects when assessing delayed tree mortality and post-fire recovery. The absence of strong correlations between PDSI and dNBR might indicate the presence of other complex factors that also influence burn severity, warranting further investigation.

Model Performance

This study evaluated three machine learning models (SVC, Random Forest, Decision Tree and K-Nearest Neighbors) to predict tree mortality based on environmental variables. The models were trained and tested on a dataset comprising water stress and burn severity indices in forests with fire-impacted and living trees identified through remote sensing techniques.

The Random Forest model emerged as the clear winner with an outstanding accuracy of 94.96%, significantly outperforming SVC (60.83%), K-Nearest Neighbors (83.09%), and even Decision Trees (90.95%). This superior performance indicates Random Forest's exceptional ability to capture complex relationships between the predictor variables (burn severity, water stress, etc.) and the target outcome (tree mortality).

Precision, Recall, and F1-Score Analysis

Random Forest: Remains the top performer with remarkably high precision (0.9494), recall (0.9496), and F1-score (0.9466), demonstrating its exceptional ability to accurately identify tree mortality while minimizing both false positives and false negatives. This translates to a robust capacity to distinguish fire-impacted trees from living trees across all classes. Decision Trees delivered a strong accuracy of 90.95% with an F1-score of 0.9104, indicating a good balance between precision (0.9124) and recall (0.9095). However, it fell slightly short of Random Forest in terms of overall accuracy and precision. KNeighbors achieved a moderate F1-score of 0.8158, suggesting a decent balance between precision (0.8148) and recall (0.8309). However, it experienced more false positives and negatives compared to both Random Forest and Decision Trees. SVC exhibited the lowest F1-score (0.5587), indicating a less balanced performance. Its precision (0.5207) and recall (0.6083) were also significantly lower than other models, suggesting a higher tendency for both false positives and false negatives.

Confusion Matrix Insights

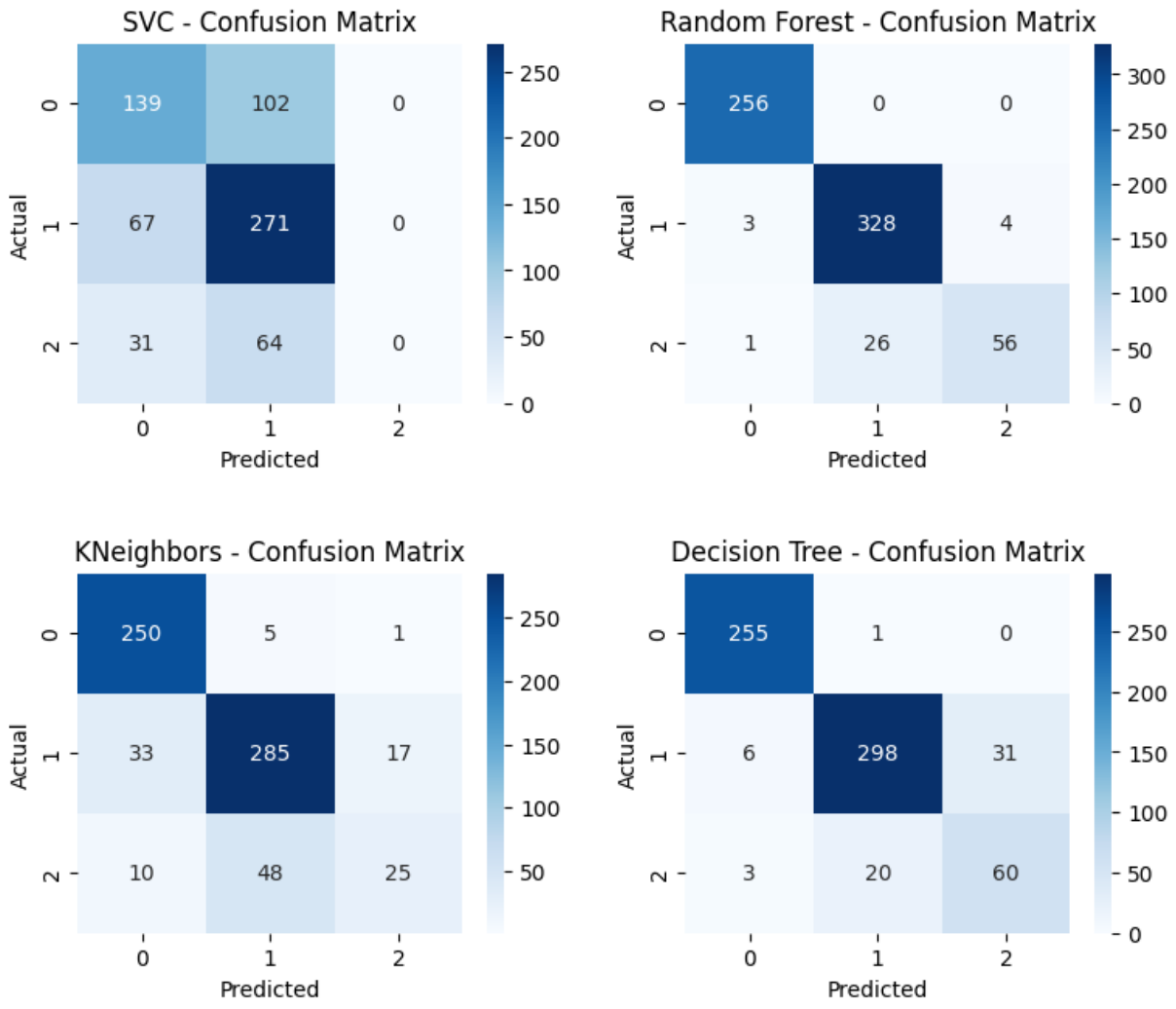


Figure 13: **Confusion Matrix for the model predictions** ***(Credit: Reva Khaire)***

Note: Here, class 0 indicates that the tree is alive, class 1 indicates that the tree died immediately after the wildfire, and class 2 indicates the tree died after one year.

The Random Forest model exhibited outstanding true positive rates for all tree mortality classes: 256 for Class 1, 328 for Class 2, and 56 for Class 3. Notably, it also minimized false positives and negatives, demonstrating exceptional ability to accurately distinguish living and fire-impacted trees across all categories. Following closely behind, Decision Trees achieved impressive true positive rates: 255 for Class 1, 298 for Class 2, and 60 for Class 3. While slightly lower than Random Forest, these numbers still indicate strong performance in tree mortality prediction. Similar to Random Forest, Decision Trees maintained low false positive and negative rates, suggesting a robust capacity for accurate classification. K-Nearest Neighbors model exhibited high true positives for the first two classes: 250 for Class 1 and 285 for Class 2. However, it struggled with Class 3, achieving only 25 true positives. This resulted in a higher number of false positives and negatives compared to the top performers, particularly for Class 3,

indicating challenges in differentiating critically stressed trees from fire-impacted trees. SVC performed poorly across all classes. It achieved only 139, 271, and 0 true positives for Classes 1, 2, and 3, respectively. This significantly lower performance, coupled with a higher rate of false positives and negatives, suggests limited success in accurately classifying tree mortality.

In conclusion: While Random Forest and Decision Trees demonstrated exceptional true positive rates and minimal misclassification, K-Nearest Neighbors exhibited strengths in the first two classes but faltered in the third. SVC, on the other hand, requires further exploration to improve its classification accuracy for reliable tree mortality prediction.

Importance of Feature Selection

The superior performance of Random Forest suggests it effectively learned complex relationships between the chosen features and tree mortality. This highlights the importance of selecting relevant features that hold predictive power for the chosen model.

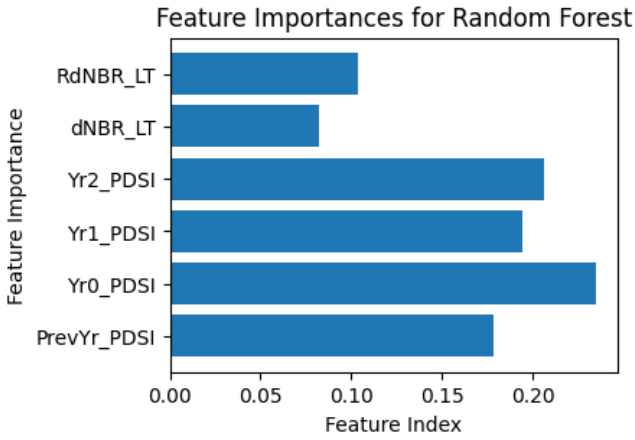


Figure 14: Future importance chart for Random Forest model (Credit: Reva Khaire)

Random forest model gives almost equal weightage to 4 years of drought history with drought in the year of fire getting the highest importance. Long term relative dNBR (RdNBR_LT) also plays an important role in random forest classification.

Generalizability and Future Work

While the results are promising, further validation on independent datasets is necessary to assess the generalizability of the Random Forest model. Additionally, exploring other model architectures and incorporating additional data sources with multiple temporal dimensions could potentially improve accuracy and refine our understanding of delayed tree mortality due to wildfires.

Conclusion

The research, utilizing the Fire and Tree Mortality (FTM) database across multiple U.S. states, provided significant insights into post-fire effects on trees. Analysis of Normalized Burn Ratio (NBR) revealed stark differences between fire-impacted and living trees, emphasizing heightened burn severity in deceased trees. Palmer Drought Severity Index (PDSI) analysis showcased increased water stress in

fire-affected trees post-fire, attributing it to disrupted hydrological cycles and amplified evapotranspiration.

Machine learning models, especially the Random Forest model, demonstrated exceptional accuracy in predicting tree mortality, highlighting the significance of feature selection in understanding complex relationships between environmental variables and tree mortality post-wildfires. Confusion Matrix insights revealed robust classification by Random Forest and Decision Trees, while SVC showed lower performance, suggesting the need for accuracy improvement.

This study underscores the critical role of factors such as drought history and burn severity in understanding post-fire tree mortality. Moreover, it showcases the effectiveness of remote sensing and machine learning in post-wildfire tree mortality prediction, suggesting its potential application in wildfire management. Further research is vital to enhance model generalizability and validate its practical applicability in informing post-wildfire forest recovery strategies, potentially minimizing economic and ecological losses.

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Appendix

Fire and Tree Mortality (FTM) database

The Fire and Tree Mortality (FTM) database is a crucial asset in contemporary fire ecology research, providing extensive records of fire events and their impacts on tree mortality (Cansler et al., 2020b). This database consolidates detailed data on fire severity, geographic and temporal specifics, and species-specific mortality rates, enabling a deeper understanding of the effects of fire on forests. Leveraging this database supports predictive modeling, aiding in the comprehension of factors influencing post-fire tree mortality and enhancing ecological assessments.

Remote Sensing Data

Research leverages remote sensing data, revolutionizing our understanding of post-fire tree mortality and drought severity. This approach expedites decision-making in post-fire impact analysis and recovery planning, aiding resource allocation and ecological restoration. The acceleration in decision-making stems from the rapid acquisition of high-resolution, spatially explicit information regarding post-fire tree mortality (using indices such as dNBR and RdNBR) and drought severity (using PDSI). Leveraging Landsat 8's multispectral data with 30-meter spatial resolution enables crucial indices' calculation for ecological assessment.

Normalized Burn Ratio (NBR)

The Normalized Burn Ratio (NBR) quantifies post-fire spectral changes in vegetation health and burn severity by contrasting near-infrared (NIR) and shortwave infrared (SWIR) bands in remote sensing data (Key & Benson, 2006).
The formula for NBR is given by:

$$NBR = 1000 . \frac{NIR+SWIR2}{NIR-SWIR2}$$

Here, Infrared Band (NIR) represents healthy vegetation and captures high reflectance due to chlorophyll absorption. Shortwave InfraredBand (SWIR2) indicates burnt or stressed vegetation and reflects low absorption due to changes in cell structure and moisture content post-fire (Key & Benson, 2006).

Difference Normalized Burn Ratio (dNBR)

dNBR is derived by computing the difference in NBR values before and after a fire event (Miller & Thode, 2007). The formula for dNBR is given by:

$$dNBR = NBR_{prefire} - NBR_{postfire}$$

Here, $NBR_{prefire}$ denotes the NBR value before the fire event, and $NBR_{postfire}$ represents the NBR value after the fire event.

Relative differenced Normalized Burn Ratio (RdNBR)

RdNBR quantifies the change in NBR concerning the minimum NBR values observed in severely burned areas (Miller & Thode, 2007). The formula for RdNBR is given by:

$$RdNBR = \frac{dNBR}{\left|(NBR_{prefire})^{0.5}\right|}$$

Here, $NBR_{prefire}$ denotes the NBR value before the fire event, and $NBR_{postfire}$ represents the NBR value after the fire event.

The assimilation of NBR data in post-fire tree mortality studies helps us understand fire-induced stress on vegetation and its resultant mortality rates. Offering a quantifiable evaluation of vegetation health post-fire, the NBR significantly contributes to unraveling the complexities associated with post-fire ecosystem recuperation.

Palmer Drought Severity Index (PDSI)

The Palmer Drought Severity Index (PDSI) assesses long-term drought severity based on precipitation, temperature, soil moisture, and other climate factors (Palmer, 1965).

Key Input Parameters used to calculate PDSI

- Precipitation (P): The amount of rainfall or snowfall received at a location, typically measured in millimeters or inches.
- Potential Evapotranspiration (PET): The amount of water that would be evaporated and transpired from a vegetated surface under ideal conditions, representing atmospheric water demand. It's often estimated using temperature, humidity, wind speed, and solar radiation data.
- Available Water Content (AWC): The capacity of the soil to store water, influenced by soil texture, structure, and organic matter content. It determines the soil's ability to buffer against drought.
- Antecedent Precipitation Index (API): A measure of antecedent moisture conditions, representing the cumulative precipitation deficit or surplus over a specified period leading up to the current time. It accounts for the persistence of drought or wetness.
- Previous PDSI: The PDSI value from the previous time step, providing a historical context for drought evolution and accounting for the cumulative effects of moisture anomalies.

PDSI values range from -4 (extremely dry) to +4 (extremely wet), with a baseline of 0 indicating near-normal conditions. Negative values signify drought conditions, while positive values indicate wet conditions. The magnitude of the value reflects drought severity or wetness intensity.

Landsat 8 Satellite Images

Landsat 8, launched on an Atlas-V rocket in 2013 from Vandenberg AFB, carries OLI & TIRS instruments for land imaging.

- OLI (Landsat's Operational Land Imager sensor): OLI measures in the visible, near infrared, and shortwave infrared portions (VNIR, NIR, and SWIR) of the spectrum.
- ETM+ (Landsat's Enhanced Thematic Mapper Plus sensor): The ETM+ contains eight spectral bands, including a pan and thermal band: Band 1 Blue (0.45 - 0.52 µm) 30 m | Band 2 Green (0.52 - 0.60 µm) 30 m | Band 3 Red (0.63 - 0.69 µm) 30 m | Band 4 Near-Infrared (0.77 - 0.90

μm) 30 m | Band 5 Short-wave Infrared (1.55 - 1.75 μm) 30 m | Band 6 Thermal (10.40 - 12.50 μm) 60 m Low Gain / High Gain | Band 7 Mid-Infrared (2.08 - 2.35 μm) 30 m | Band 8 Panchromatic (PAN) (0.52 - 0.90 μm) 15 m

- TM (Landsat's Thematic Mapper sensor): The TM contains seven spectral bands, including a thermal band: Band 1 Visible Blue (0.45 - 0.52 μm) 30 m | Band 2 Visible Green (0.52 - 0.60 μm) 30 m | Band 3 Visible Red (0.63 - 0.69 μm) 30 m | Band 4 Near-Infrared (0.76 - 0.90 μm) 30 m | Band 5 Near-Infrared (1.55 - 1.75 μm) 30 m | Band 6 Thermal (10.40 - 12.50 μm) 120 m | Band 7 Mid-Infrared (2.08 - 2.35 μm) 30 m | Ground Sampling Interval (pixel size): 30 m reflective, 120 m thermal

Machine learning Classification Models used in this research

- **Support Vector Classifier (SVC):** Utilizes support vectors to create decision boundaries in a high-dimensional space, aiming to separate different classes effectively. Particularly useful for binary and multiclass classification, it identifies the hyperplane that maximizes the margin between classes. Through a kernel trick, it can handle complex, nonlinear relationships in the data. Its strengths lie in its ability to handle high-dimensional data and its robustness in capturing intricate decision boundaries.

- **Decision Tree**: Operates through recursive partitioning, systematically dividing the dataset into subsets according to feature values, constructing a tree-shaped structure in the process. It relies on metrics like Gini Impurity or Information Gain to discern the most suitable feature and split point at each node, ensuring optimal partitioning. Being non-parametric, this method refrains from presumptions about the inherent data distribution, allowing it to adapt flexibly to various data types.

- **Random Forest**: Operates as an ensemble method by amalgamating numerous decision trees, thereby enhancing accuracy and mitigating overfitting concerns. Employing bagging and feature subsampling techniques, each tree within the ensemble is trained on distinct random subsets of both the data and features, fostering diversity in predictions. The final prediction is derived from a consensus, employing a majority vote for classification or an average for regression, based on the individual tree predictions. Random Forest demonstrates robustness by exhibiting reduced sensitivity to noise and outliers when contrasted with single decision trees.

- **K-Nearest Neighbors (KNN)**: Operates under a principle of lazy learning, preserving all training data until a new query point triggers the classification process. Utilizing various distance metrics like Euclidean or Manhattan distances, it identifies the K closest neighbors to the given query point. The classification decision is then made through a majority vote mechanism, determining the class label based on the most frequent occurrence among the K nearest neighbors. An essential hyperparameter in KNN involves selecting the value of K, a choice that significantly influences the model's performance, as it dictates the number of neighbors considered during classification.